

AR 527: Patenting through Design Thinking

Course Code	AR 527
Course Name	Patenting through Design Thinking
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	PG, and UG 3rd and 4th Year Students
Approval	61st BoA (Jan 2026); 49th Senate (Jun 2026), in the Minor in Entrepreneurship curriculum

Preamble

This Course starts with an overview of the process and how design thinking helps individuals or organizations. Initially, we make a high-level introduction of the concepts so that the terminology used is seamlessly understood through a 3-circle diagram, identify key traits and values associated with design thinking methodology, stages of design thinking, benefits, and characteristics of a well-designed solution. We also talk about how one can evaluate problems and create a team (of employees or students) to help one solve that problem. Then we explore ways to understand the users, evolve innovative ideas, and test the inferences to ensure user satisfaction, for a greater understanding of design thinking. Design thinking can enhance patentability and commercial viability. It can also enhance an individual or Institution to ensure that a greater percentage of filed patents get granted and also it can help one ensure that a granted patent can fetch adequate revenue through technology transfer/licensing or entrepreneurship. The goal of this course is to achieve patenting through design thinking. The learning objectives for students include the ability to draft patents and apply design thinking stages to IP creation and/or apply strategies to commercialize patents.

Course Modules

- **Unit 1:** What is Design Thinking? The triad of Business, User and Technology; The Research, Ideation, and Prototyping stages of Design Thinking. Difference between invention and design; Innovation, invention, and creativity; Technology Readiness Levels (TRLs) and Business readiness levels (BRLs). (10 Hours)
- **Unit 2:** Patents: Introduction, The Turmeric Story, Edison's Dictum, Prior Art Search, 'What can and cannot be patented?' Patenting landscape: number of filed patents, granted and commercialized, and why the conversion to commercialization is low; Provisional and Complete Specifications; First-to-File Concept; Infringement, Government support system for patenting and commercialization: various financial and non-financial support available. (10 Hours)
- **Unit 3:** Patent Strategies for Startups, Academia, and MSMEs; Patent filing tactics and managing the patent life cycle, Design around, Utilization of Patents: Exclusive and Non-exclusive Patent Licensing, Technology Transfer, Spinoff, Startup, Productization etc. (12 Hours)
- **Unit 4:** Evaluating the cost-effectiveness of patenting and maintaining a patent, Patent drafting, Patenting through design thinking, Mock Patent filing. (10 Hours)

Laboratory Modules

None.

Text Books

1. Kankanala, K.C. *Indian patent law and Practice*. New Delhi: OUP India, 2014.
2. Brown, T. and Katz, B. *Change by design: How design thinking can transform organizations and inspire innovation*. New York, NY: HarperCollins Publishers, 2009.

References

1. United States Patent and Trademark Office (<https://www.uspto.gov>)
2. India Patent Office (<https://www.ipindia.gov.in>)